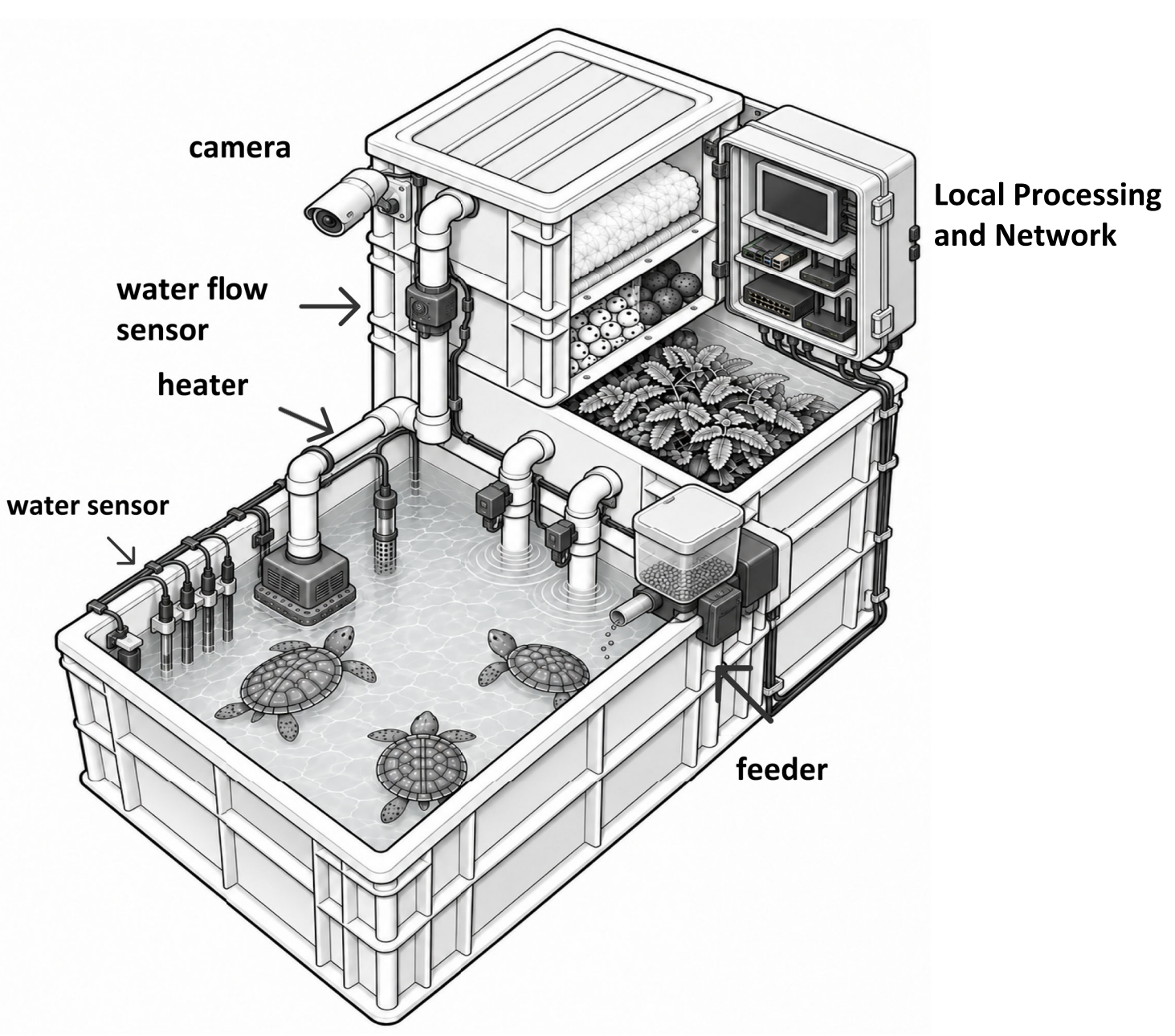
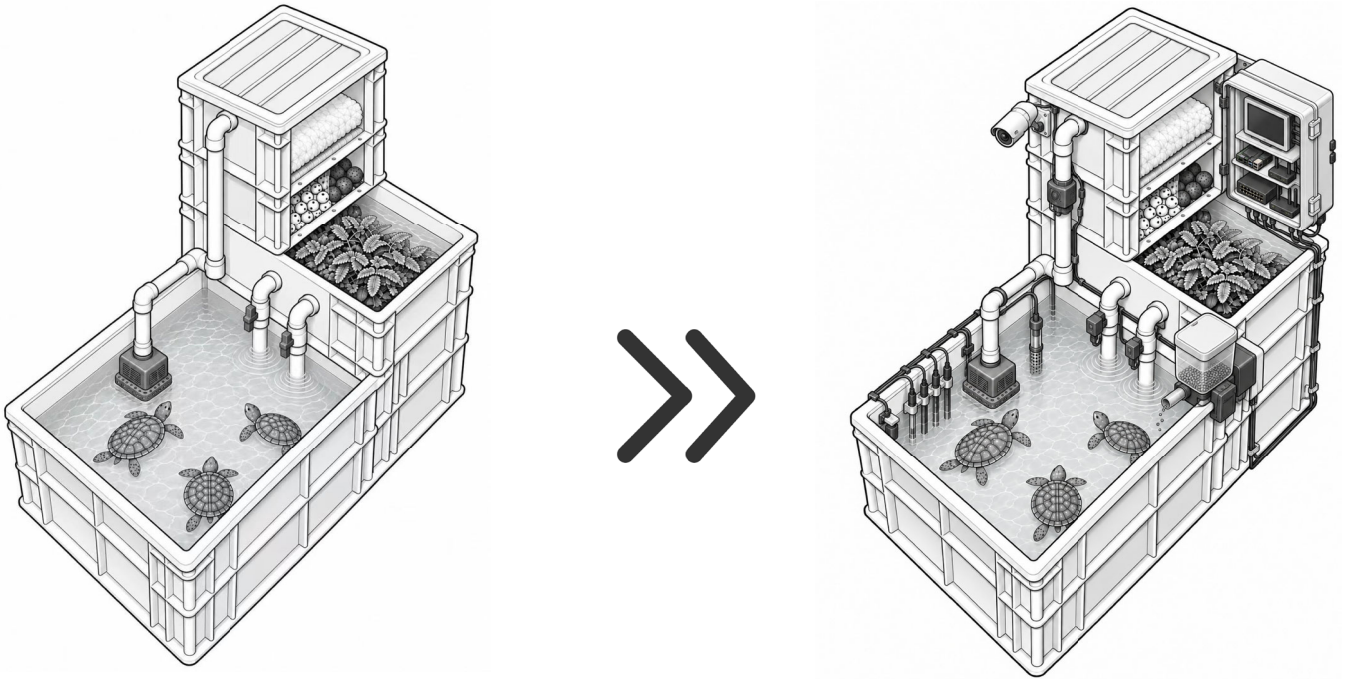
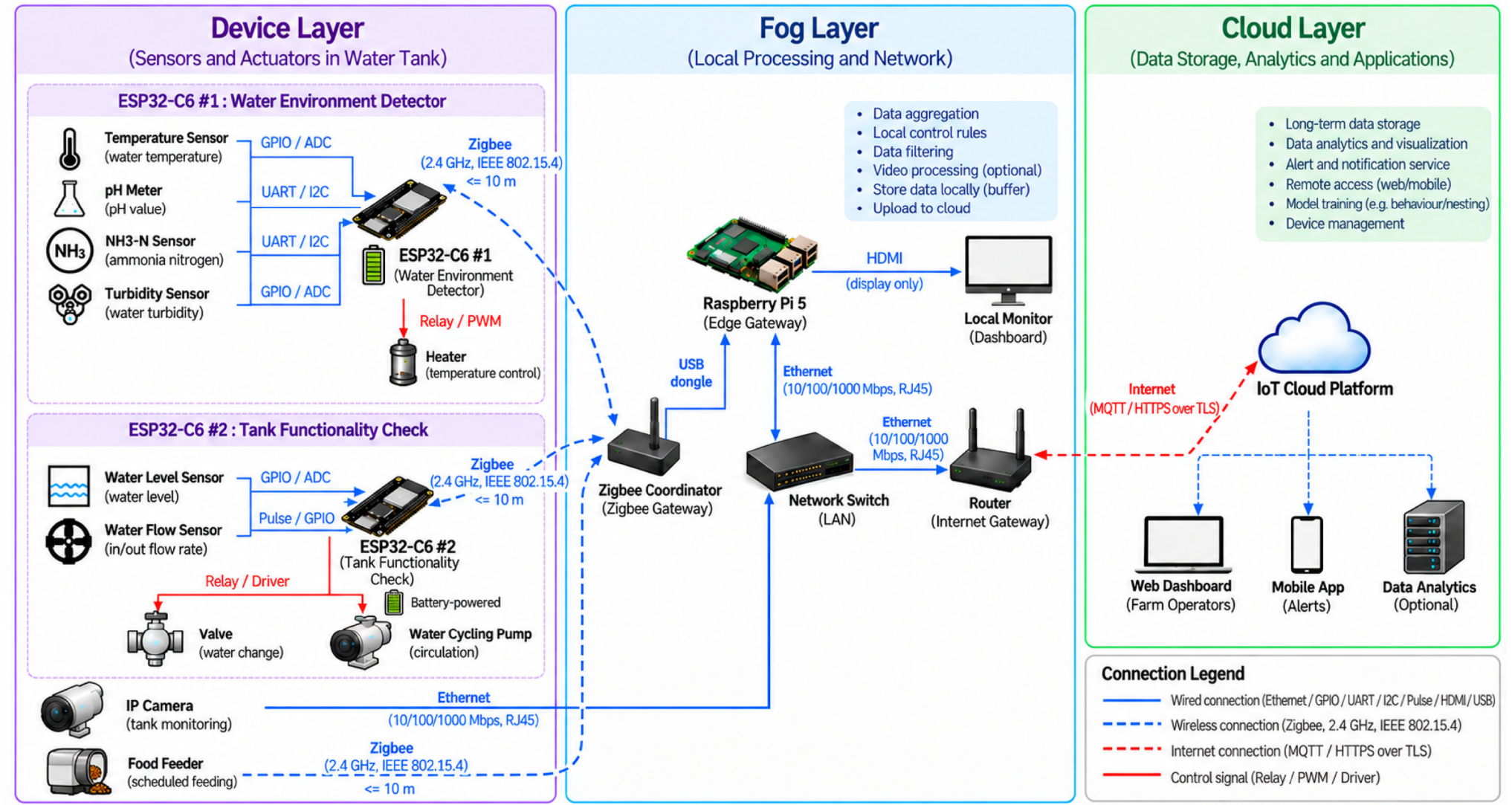
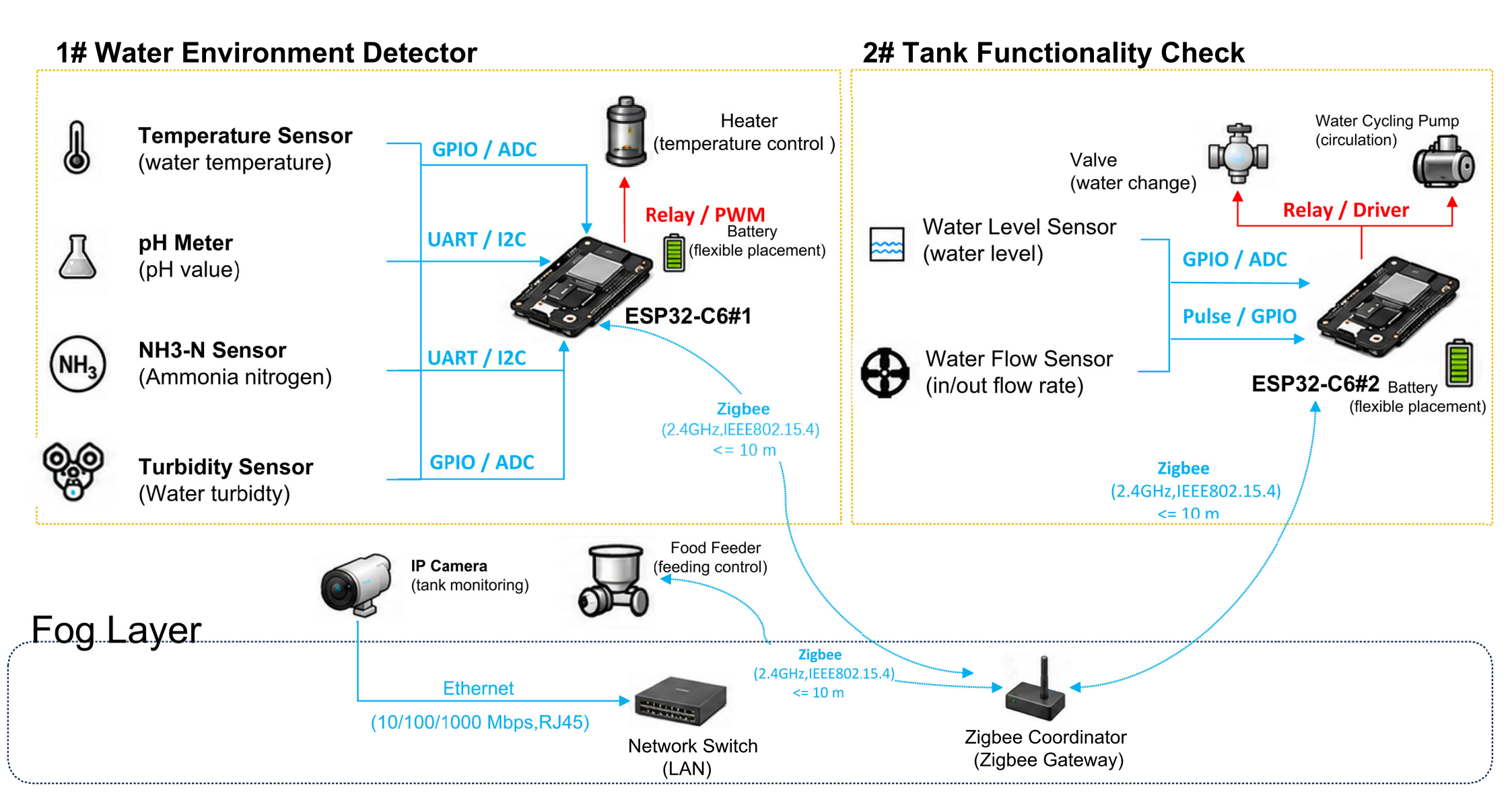
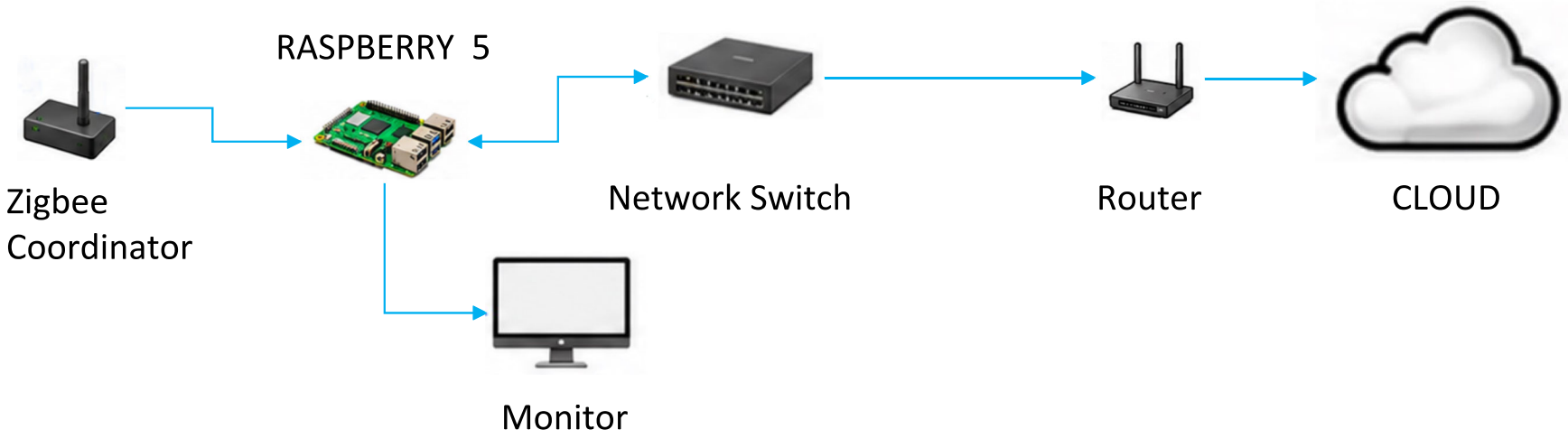
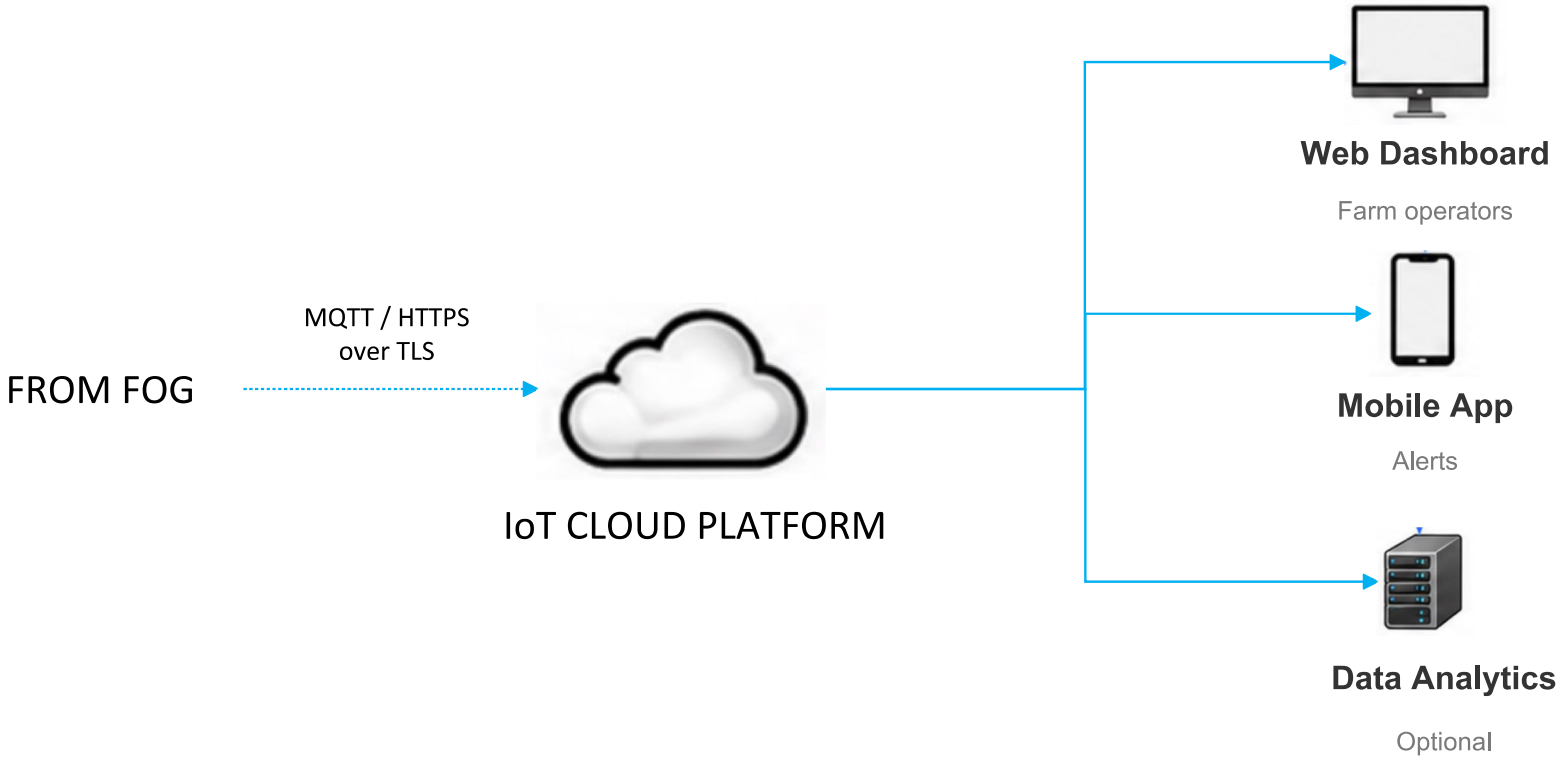

camera
Local Processing and Network
water flow sensor
heater
water sensor
feeder




Device Layer
(Sensors and Actuators in Water Tank)
ESP32-C6 #1 : Water Environment Detector
Temperature Sensor
(water temperature)
pH Meter
(pH value)
NH3-N Sensor
(ammonia nitrogen)
Turbidity Sensor
(water turbidity)
GPIO / ADC
UART / I2C
UART / I2C
GPIO / ADC
Zigbee
(2.4 GHz, IEEE 802.15.4)
<= 10 m
ESP32-C6 #1
(Water Environment Detector)
Relay / PWM
Heater
(temperature control)
ESP32-C6 #2 : Tank Functionality Check
Water Level Sensor
(water level)
Water Flow Sensor
(in/out flow rate)
GPIO / ADC
Pulse / GPIO
Zigbee
(2.4 GHz, IEEE 802.15.4)
<= 10 m
ESP32-C6 #2
(Tank Functionality Check)
Relay / Driver
Battery-powered
Valve
(water change)
Water Cycling Pump
(circulation)
IP Camera
(tank monitoring)
Ethernet
(10/100/1000 Mbps, RJ45)
Food Feeder
(scheduled feeding)
Zigbee
(2.4 GHz, IEEE 802.15.4)
<= 10 m
Fog Layer
(Local Processing and Network)
Data aggregation
Local control rules
Data filtering
Video processing (optional)
Store data locally (buffer)
Upload to cloud
Raspberry Pi 5
(Edge Gateway)
HDMI
(display only)
Local Monitor
(Dashboard)
USB dongle
Ethernet
(10/100/1000 Mbps, RJ45)
Zigbee Coordinator
(Zigbee Gateway)
Network Switch
(LAN)
Ethernet
(10/100/1000 Mbps, RJ45)
Router
(Internet Gateway)
Cloud Layer
(Data Storage, Analytics and Applications)
Long-term data storage
Data analytics and visualization
Alert and notification service
Remote access (web/mobile)
Model training (e.g. behaviour/nesting)
Device management
Internet
(MQTT / HTTPS over TLS)
IoT Cloud Platform
Web Dashboard
(Farm Operators)
Mobile App
(Alerts)
Data Analytics
(Optional)
Connection Legend
Wired connection (Ethernet / GPIO / UART / I2C / Pulse / HDMI / USB)
Wireless connection (Zigbee, 2.4 GHz, IEEE 802.15.4)
Internet connection (MQTT / HTTPS over TLS)
Control signal (Relay / PWM / Driver)


1# Water Environment Detector
2# Tank Functionality Check
Temperature Sensor
(water temperature)
pH Meter
(pH value)
NH3-N Sensor
(Ammonia nitrogen)
Turbidity Sensor
(Water turbidty)
GPIO / ADC
UART / I2C
UART / I2C
GPIO / ADC
Heater
(temperature control )
Relay / PWM
Battery
(flexible placement)
ESP32-C6#1
Zigbee
(2.4GHz,IEEE802.15.4)
<= 10 m
Valve
(water change)
Water Cycling Pump
(circulation)
Relay / Driver
Water Level Sensor
(water level)
Water Flow Sensor
(in/out flow rate)
GPIO / ADC
Pulse / GPIO
ESP32-C6#2
Battery
(flexible placement)
Zigbee
(2.4GHz,IEEE802.15.4)
<= 10 m
IP Camera
(tank monitoring)
Food Feeder
(feeding control)
Fog Layer
Ethernet
(10/100/1000 Mbps,RJ45)
Network Switch
(LAN)
Zigbee
(2.4GHz,IEEE802.15.4)
<= 10 m
Zigbee Coordinator
(Zigbee Gateway)


RASPBERRY 5
Zigbee
Coordinator
Network Switch
Router
CLOUD
Monitor


FROM FOG
MQTT / HTTPS
over TLS
IoT CLOUD PLATFORM
Web Dashboard
Farm operators
Mobile App
Alerts
Data Analytics
Optional